



# Percolation on Fractal Networks: A Survey

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**Abstract:** The purpose of this survey is twofold. First, we survey the studies of percolation on fractal networks. The objective is to assess the current state of the art on this topic, emphasizing the main findings, ideas and gaps in our understanding. Secondly, we try to offer guidelines for future research. In particular, we focus on effects of fractal attributes on the percolation in self-similar networks. Some challenging questions are outlined.

**Keywords:** percolation; fractal networks; dimension numbers; percolation threshold; critical exponents; universality classes

## 1. Introduction

Percolation phenomena are ubiquitous in nature [1–3]. Although the notion of percolation dates back to studies of gelation in polymers performed by Paul Flory in early 1940s (see references in Ref. [2]), the first mathematical model of percolation was developed by Broadbent and Hammersley in 1957 [4]. Nowadays, the percolation is a paradigmatic model for probability theory with a wealth of scientific and engineering applications [5–7]. A classical site percolation problem is defined on a translation invariant network. Each network site can be either occupied with probability  $p$ , or empty with the probability  $1 - p$ . In the case of Bernoulli percolation, the occupancy of different sites is uncorrelated. Nonetheless, neighboring occupied sites merge to form clusters. When the occupation probability is low, the occupied sites either are isolated, or form very small clusters. However, for sufficiently large  $p$ , a lot of occupied sites form one large cluster that can reach two opposite sides of the lattice (e.g., top or /and bottom). The lowest concentration of occupied sites for which there is a spanning or percolating cluster occupying a finite fraction of the total number of sites on an infinite lattice,  $p_C$ , is called the site percolation threshold [8]. Likewise, the bond percolation threshold is defined as the lowest concentration of connected bonds at which there is the percolation cluster.

The sudden onset of the spanning cluster at  $p_C$  is accompanied by scaling behaviors of system properties characterized by a set of critical exponents [9]. A geometric phase transition at the percolation threshold constitutes a particularly elegant example of second order phase transition [10–13]. Despite a purely geometric nature, the percolation transition embodies many of the key aspects of critical phenomena. In this regard, it has been recognized that the percolation transition can be treated as a particular case of the  $q$ -state Potts model with  $q$  being equal to 1 [14,15]. The critical exponents in Euclidean system depend upon the space dimension and not on the choice of particular model [10–16].

Since in fractals the translational symmetry is broken, the fundamental question is how the scale invariance affects the critical phenomena on the fractal networks. This question was first raised by Gefen et al. [17], in their study on the effects of fractional dimensionality on the critical phenomena on fractals. In this regard, it was recognized that the critical properties are dependent not only on the fractal dimension of the network, but also on other topological attributes [17–21]. In particular, with an approximate renormalization group recursion relation, Gefen et al. [20] found that a finitely ramified Sierpinski gasket has



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only trivial percolation threshold,  $p_C = 1$ , for both the site and the bond percolation. This result was reproduced in [22] by other means. Later, this finding was confirmed with exact renormalization group recursion relations for the finitely ramified Sierpinski gaskets and branching Koch curves [23,24]. On the other hand, Trugman and Weinrib [25] proposed a model of percolation with a threshold at zero: the network conducts no matter how small the volume fraction of conductor. The random scale-free networks with the power-law degree distribution exponent lying between 2 and 3 also have the trivial percolation threshold  $p_C = 0$ , meaning that networks are always in the percolated phase [26]. Later, it was proved that self-similar networks possessing a hierarchy of nested subgraphs whose average degree grows with their depth in the hierarchy also have  $p_C = 0$  [27]. Percolation in hierarchical scale-free networks was studied in Ref. [28]. Depending on the method of construction, the hierarchical network can be fractal or small world, assortative or disassortative, or possess various degrees of clustering. Accordingly, different types of criticality were found, illustrating the crucial effect of other structural properties aside from the scale-free degree distribution.

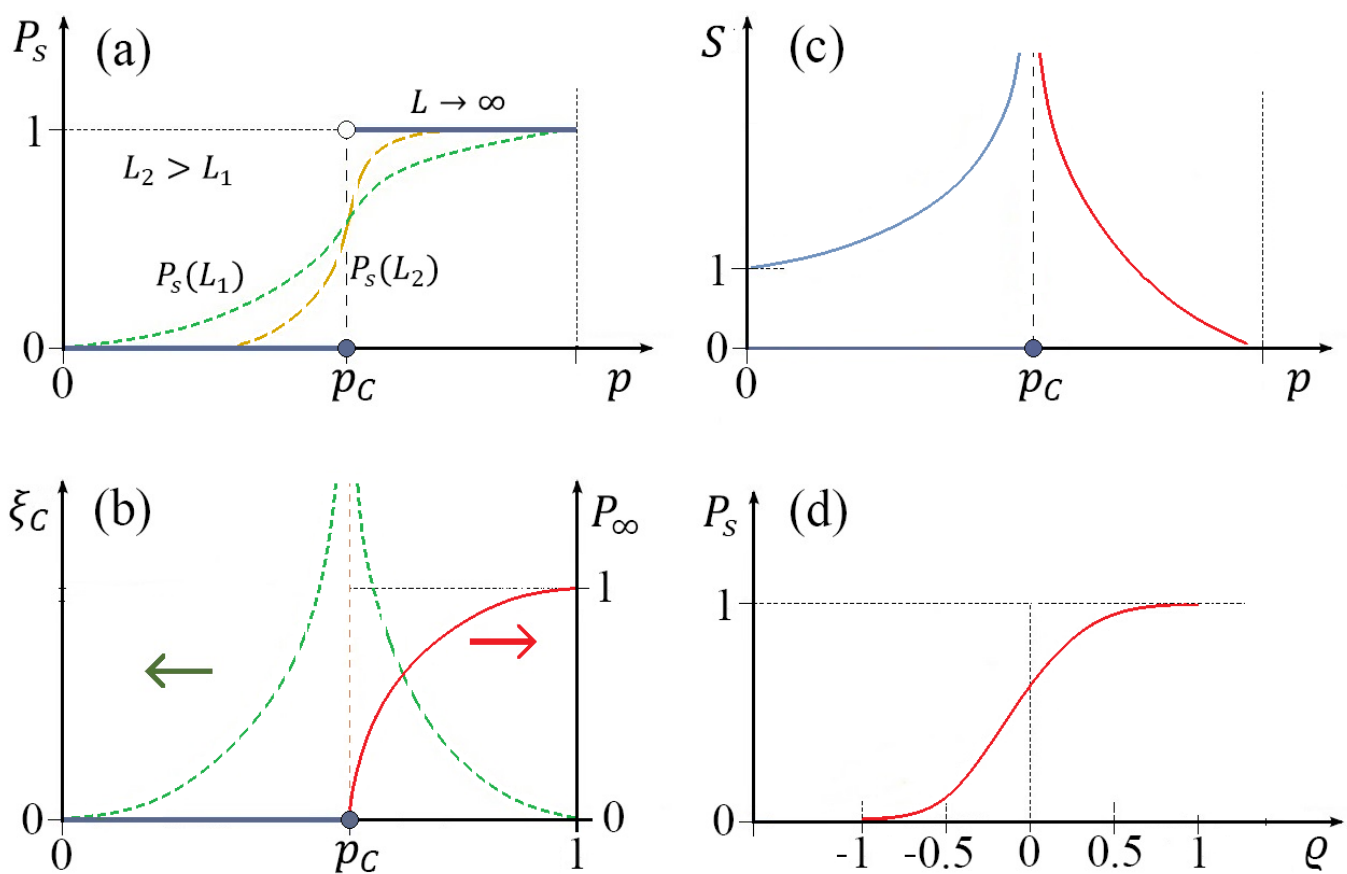
The first numerical evidence of percolation phase transition on infinitely ramified fractals was presented by Havlin et al. in 1983 [29]. Further, it was found that the critical exponents for percolation are strongly dependent on the topology and geometry of the fractal network [29–31]. In Ref. [32], the site percolation thresholds for infinitely ramified Sierpinski carpets of different lacunarity were estimated using the translational-dilation method and Monte Carlo technique. Percolation thresholds and universality of site percolation on infinitely ramified Sierpinski carpets were also studied by authors of Ref. [33]. The critical behavior of bond percolation on infinitely ramified Sierpinski carpets was studied in Ref. [34] by the renormalization group approach. In [35], the site percolation transition in random Sierpinski carpets was studied by real space renormalization. The statistical geometry of percolation clusters was studied in [36]. Sukop et al. [37] performed a study of percolation thresholds in random fractals used as models of porous media. The site and bond percolation on multifractal scale-free planar stochastic lattices were studied in Refs. [38–41]. Herega et al. [42] studied percolation on hybrid-ramified Sierpinski carpets. Wu [43] has proved the uniqueness of the critical percolation cluster on the Sierpinski carpet. Correlated rigidity percolation in fractal lattices was studied in [44]. Recursive percolation on fractal percolation clusters was studied in [45]. Against this background, Shinoda [46] has argued that only infinitely ramified networks can have non-trivial percolation thresholds. The first systematic study of percolation on infinitely ramified Sierpinski carpets with different fractal dimensions was performed by Monceau and Hsiao [47]. Later, Higuchi and Wu [48] have proved the uniqueness of the percolation threshold for the Sierpinski carpet lattices. Comprehensive studies of site percolation on infinitely ramified Sierpinski carpets were reported in Refs. [49,50]. The effects of network attributes on the percolation threshold and values of critical exponents were discussed in Refs. [47–51]. It has been recognized that the fractional dimension alone is insufficient to characterize the fractal features affecting the critical exponents. However, the effects of fractal geometry on the network percolation are not completely understood.

Our goal in this work is to give an overview of research works on percolation in fractal networks. The rest of the paper is organized as follows. The main characteristics of percolation phenomena are sketched in Section 2. In Section 3, the key attributes of fractal networks are scrutinized. Percolation on finitely ramified fractals is briefly discussed in Section 4. Section 5 is devoted to the studies of percolation on infinitely ramified fractal networks. In particular, we focus on effects of the fractal attributes on the percolation features. The challenging problems and possible guidelines are outlined in Section 6.

## 2. Main Features of Percolation

The classic percolation theory deals with the clustering properties of occupied sites (connected bonds) which are randomly and uniformly distributed in the network with a finite occupation probability  $p$ . A key feature of percolation is the cluster connectedness

associated with spanning probability  $P_s(p, L)$  which depends on the system size  $L$ , as well as on the occupancy probability  $p$  (see Figure 1a). In the limit of an infinite lattice ( $L \rightarrow \infty$ ), there exists a well-defined threshold probability  $p_C$ , above which an infinitely large cluster spans the system. As long as  $p < p_C$ , all clusters are finite and their size distribution has a tail which decays exponentially. Conversely, if  $p > p_C$ , there exists an infinite cluster with probability one, while the size distribution of other (finite) clusters has a tail which decays slower than exponentially. Hence, at  $p = p_C$ , the system undergoes a continuous phase transition between the globally connected and disconnected phases (see Figure 1b–d). The values of percolation thresholds for the Bernoulli percolation on translationally invariant lattices and random networks are summarized in Table 1. An elegant generalization of Bernoulli percolation model was proposed by Fortuin and Kasteleyn [16]. Later it was recognized that the Fortuin–Kasteleyn percolation (also called the random-cluster model) is related to many other models of statistical mechanics, including the Ising and Potts models, while the Bernoulli percolation, if recovered in the limit of the cluster-weight,  $q = 1$  [14].



**Figure 1.** Schematic plot of the: (a) percolation probability  $P_s(p, L)$ ; (b) correlation length  $\xi_C$  and percolation strength  $P_\infty$ ; and (c) mean size of finite clusters  $S$  as functions of the occupancy  $p$  for a system of infinite size  $L$  (as the system size  $L$  goes to infinity, the percolation probability  $P_s$  tends to the step function around the critical threshold  $p_C$ ); (d) collapse of the percolation (spanning) probabilities in coordinates  $P_s(p, L)$  versus  $q = (p - p_C)L^{1/\nu}$ .

**Table 1.** Percolation thresholds for translationally invariant lattices and random networks.

| $d$                                                  | Network                            | $\langle k \rangle$ | Percolation Threshold |                                            |
|------------------------------------------------------|------------------------------------|---------------------|-----------------------|--------------------------------------------|
|                                                      |                                    |                     | Site Percolation      | Bond Percolation                           |
| 1                                                    | Chain [52]                         | 2                   | 1                     | 1                                          |
| 2                                                    | WPSL [39,40]                       | 5.333               | 0.5265                | 0.3457                                     |
|                                                      | Martini lattice [53,54]            | 3                   | 0.764826 ...          | $1/\sqrt{2} = 0.70710678 \dots$            |
|                                                      | Planar $\Phi^3$ random graphs [55] | 3                   | 0.7360 ...            | -                                          |
|                                                      | Voronoi network [56]               | 3                   | 0.71410 ...           | 0.666931                                   |
|                                                      | Delaunay network [56]              | 6                   | 1/2                   | 0.333069 ...                               |
|                                                      | Honeycomb [57]                     | 3                   | 0.697043 ...          | $1 - 2 \sin(\pi/18) \approx 0.65271 \dots$ |
|                                                      | Square [57]                        | 4                   | 0.592746 ...          | 1/2                                        |
|                                                      | Triangular [57]                    | 6                   | 1/2                   | $2 \sin(\pi/18) \approx 0.34729 \dots$     |
|                                                      | Kagomé [58]                        | 4                   | $1 - 2 \sin(\pi/18)$  | 0.5244053 ...                              |
|                                                      | Bowtie [59]                        | 5                   | 0.5474 ...            | -                                          |
| 3<br>[60–62]                                         | Diamond                            | 4                   | 0.4301 ...            | 0.3893 ...                                 |
|                                                      | Simple cubic                       | 6                   | 0.3116077 ...         | 0.24881182 ...                             |
|                                                      | Body-centred cubic                 | 8                   | 0.2459615 ...         | 0.1802875 ...                              |
|                                                      | Face-centred cubic                 | 12                  | 0.1992365 ...         | 0.1201635 ...                              |
|                                                      | Hexagonal close packed             | 12                  | 0.1992555 ...         | 0.1201640 ...                              |
| 4                                                    | Hypercubic lattices [63]           | 8                   | 0.19688561 ...        | 0.16013122 ...                             |
| 5                                                    |                                    | 10                  | 0.14079633 ...        | 0.11817145 ...                             |
| 6                                                    |                                    | 12                  | 0.109016661 ...       | 0.09420165 ...                             |
| 7                                                    |                                    | 14                  | 0.08895112 ...        | 0.078675230 ...                            |
| $\infty$                                             |                                    | $2d$                | $\propto 1/(2d - 1)$  | $\propto 1/(2d - 1)$                       |
| $\infty$                                             | Bethe lattices [56]                | $z$                 | $\approx 1/(z - 1)$   | $\approx 1/(z - 1)$                        |
| Erdős-Rényi networks<br>$P(k) = e^{-z} z^k / k!$ [3] |                                    | $z$                 | $1/z$                 | $1/z$                                      |

One of the most striking features of percolation systems is their universal properties in the vicinity of the percolation threshold associated with the divergence of the correlation length  $\xi(p)$  as  $p \rightarrow p_C$  (see Figure 1b). In a continuous phase transition, natural observables decay algebraically. Accordingly, the criticality is characterized by a set of critical exponents which describe how the properties of the model are dependent on the length scale at which the system is observed. These exponents are quantities of primary interest in physics and have been the object of numerous studies (see, for review, Refs. [9–15] and references therein). The definitions of the percolation critical exponents are summarized in Table 2, Refs. [5–10]. It should be emphasized that the critical exponents are not entirely independent of each other but can be associated by so-called scaling relations (see Table 3, Ref. [10]). A remarkable feature of these relations is that they hold for different universality classes, meaning that the critical exponents may be different for different models, yet they always obey the same scaling relations. The universality hypothesis states that there is only a small number of distinct classes of continuous phase transitions. These classes are universal in the sense that they only depend on the dimensionality of the system and system symmetries but are independent of the interaction details of the systems. The values of critical exponents for some classical models are summarized in Table 4.

**Table 2.** Definition of the percolation critical exponents.

| Property                                                                                                                | Critical Exponent               | Scaling Relation                             |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------------------------|
| The finite cluster-size distribution<br>$n_s(p = p_C)$                                                                  | $\tau$<br>Fisher exponent       | $n_s \propto s^{-\tau}$                      |
| Moments of the cluster size distribution<br>$M_k = \sum_{s=1}^{\infty} s^k n_s$ , where $k = 0, 1, 2, \dots$            | $\sigma, \tau$                  | $M_k \propto  p - p_C ^{-(1+k-\tau)/\sigma}$ |
| The total number of finite clusters $\mathcal{N} = M_0$                                                                 | $\alpha$                        | $\mathcal{N} \propto  p - p_C ^{2-\alpha}$   |
| Percolation strength (probability that an occupied site or bond belongs to the spanning cluster) $P_{\infty} = p - M_1$ | $\beta$                         | $P_{\infty} \propto  p - p_C ^{-\beta}$      |
| Percolation susceptibility (mean size of finite clusters)<br>$S = M_2/M_1$                                              | $\gamma$                        | $S \propto  p - p_C ^{-\gamma}$              |
| Cluster moments ratio $\mathcal{M} = M_{k+1}/M_k$ for<br>$k \geq 2$                                                     | $\Delta$                        | $\mathcal{M} \propto  p - p_C ^{-\Delta}$    |
| Correlation length (spanning cluster size near the percolation threshold)                                               | $\nu$                           | $\xi_C \propto  p - p_C ^{-\nu}$             |
| The size of the largest cluster ( $s_{\max} \propto \xi_C^D$ )                                                          | $\sigma$                        | $s_{\max} \propto  p - p_C ^{-1/\sigma}$     |
| Electrical resistivity ( $\rho$ )                                                                                       | $\mu$<br>Resistivity exponent   | $\rho \propto  p - p_C ^{-\mu}$              |
| Pair-connectedness $g_c$ (the probability that two sites separated by a distance $r$ belong to the same cluster)        | $\eta$<br>Connectivity exponent | $g_c(r) \propto r^{2-d-\eta}$                |
| Probability $P_{\geq L}(p_C) = \text{Prob}[ C  \geq L]$                                                                 | $\delta$<br>Volume exponent     | $P_{\geq L}(p_C) \propto L^{-1/\delta}$      |
| Mass (number of occupied sites or bonds) of the spanning clusters                                                       | $D$<br>Fractal dimension        | $\mathcal{M}(p = p_C) \propto L^D$           |

**Table 3.** Scaling and hyperscaling relations.

|          | $\alpha, \beta$                         | $\beta, \nu$              | $\tau, \sigma$               |
|----------|-----------------------------------------|---------------------------|------------------------------|
| $\tau$   | $1 + (2 - \alpha)/(2 - \alpha - \beta)$ | $1 + d\nu/(d\nu - \beta)$ | $\tau$                       |
| $\sigma$ | $1/(2 - \alpha - \beta)$                | $1/(d\nu - \beta)$        | $\sigma$                     |
| $\alpha$ | $\alpha$                                | $2 - d\nu$                | $2 - (\tau - 1)/\sigma$      |
| $\beta$  | $\beta$                                 | $\beta$                   | $(\tau - 2)/\sigma$          |
| $\gamma$ | $2(1 - \beta) - \alpha$                 | $d\nu - 2\beta$           | $(3 - \tau)/\sigma$          |
| $\Delta$ | $2 - \alpha - \beta$                    | $d\nu - \beta$            | $1/\sigma$                   |
| $\nu$    | $(2 - \alpha)/d$                        | $\nu$                     | $(\tau - 1)/d\sigma$         |
| $\eta$   | $\beta/(2 - \alpha)$                    | $\beta/d\nu$              | $2 - d(3 - \tau)/(\tau - 1)$ |
| $\delta$ | $(2 - \alpha)/\beta - 1$                | $d\nu/\beta - 1$          | $1/(\tau - 2)$               |
| $D$      | $d - d\beta/(2 - \alpha)$               | $d - \beta/\nu$           | $d/(\tau - 1)$               |

**Table 4.** The values of critical exponent for some universality classes.

| Model                                             |                                        | $\alpha$        | $\beta$                     | $\nu$    | $\sigma$                     | $\tau$                        | $D$                          |
|---------------------------------------------------|----------------------------------------|-----------------|-----------------------------|----------|------------------------------|-------------------------------|------------------------------|
| Bernoulli percolation<br>(BP) on regular lattices | $d = 1$ [64]                           | 1               | 0                           | 1        | 1                            | 2                             | 1                            |
|                                                   | $d = 2$ [65]                           | $-2/3$          | $5/36$<br>$\approx 0.1389$  | $4/3$    | $36/91$<br>$\approx 0.3956$  | $187/91$<br>$\approx 2.0549$  | $48/91$<br>$\approx 1.8958$  |
|                                                   | $d = 3$ [66]                           | $-0.616$        | 0.405                       | 0.872    | 0.452                        | 2.190                         | 2.52293                      |
|                                                   | $d = 4$ [66]                           | $-0.712$        | 0.639                       | 0.678    | 0.482                        | 2.313                         | 3.0446                       |
|                                                   | $d = 5$ [66]                           | $-0.855$        | 0.845                       | 0.571    | 0.497                        | 2.412                         | 3.5260                       |
|                                                   | $d = 6$ [7]                            | $-1$            | 1                           | $1/2$    | $1/2$                        | $5/2$                         | 4                            |
| BP on planar random graphs $\Phi^3$ [55]          |                                        | $-2$            | $1/2$                       | 1        | $2/7$<br>$\approx 0.2857$    | $15/7$<br>$\approx 2.1428$    | $3/2$                        |
| BP on WPSL $d = 2$ [40]                           |                                        | $-1.272$        | 0.222                       | 1.635    | 0.328                        | 2.0728                        | 1.8642                       |
| Explosive percolation on WPSL $d = 2$ [41]        |                                        | $-0.272$        | 0.0679                      | 1.136    | 0.454                        | 2.03                          | 1.941                        |
| Fortuin-Kasteleyn<br>percolation<br>$n = 2$ [67]  | $q = 1$                                | $-2/3$          | $5/36$                      | $4/3$    | $36/91$                      | $187/91$                      | $91/48$                      |
|                                                   | $q = 2$                                | 0               | $1/8$                       | 1        | $8/15$                       | $31/15$                       | $15/8$                       |
|                                                   | $q = 3$                                | $1/3$           | $1/9$                       | $5/6$    | $9/14$                       | $29/14$                       | $28/15$                      |
|                                                   | $q = 4$                                | $2/3$           | $1/12$<br>$\approx 0.0833$  | $2/3$    | $4/5$                        | $31/15$<br>$\approx 2.0667$   | $15/8$                       |
| No-exclaves<br>percolation                        | $d = 2$<br>[68]                        | $-2/3$          | 0                           | $4/3$    | $3/8$                        | 2                             | 2                            |
| Minesweeper<br>percolation [69]                   | $d = 2$                                | $-0.5$          | 0.259                       | 1.25     | 0.505                        | 2.261                         | 1.7928                       |
|                                                   | $d = 3$                                | 0.42            | 0.702                       | 0.79     | 5.651                        | 9.928                         | 2.1114                       |
| Ising percolation $d = 2$ [70–72]                 |                                        | 0               | $5/96$<br>$\approx 0.05208$ | 1        | $96/187$<br>$\approx 0.5134$ | $379/187$<br>$\approx 2.0267$ | $187/96$<br>$\approx 1.9479$ |
| Ising<br>model                                    | $d = 2$ [73]                           | 0               | $1/8$                       | 1        | $8/15$                       | $31/15$                       | $15/8$                       |
|                                                   | $d = 3$ [74]                           | 0.1102          | 0.32630                     | 0.629912 | 0.63962                      | 2.2087                        | 1.482                        |
|                                                   | $d = 4$ [75]                           | 0               | $1/2$                       | $1/2$    | $2/3$                        | $7/3$                         | 3                            |
| $d = 3$                                           | Random-field Ising [76]                | $-0.16$         | 0.019                       | 1.38     | 0.243                        | 2.005                         | 2.9862                       |
|                                                   | Random-field Potts<br>( $q = 3$ ) [77] | $-0.088$        | 0.0440                      | 1.376    | 0.245                        | 2.011                         | 2.968                        |
|                                                   | Heisenberg model [78]                  | $-0.14$         | 0.362                       | 0.704    | 0.571                        | 2.207                         | 2.486                        |
|                                                   | XY model [79]                          | $-0.015$        | 0.3485                      | 0.67155  | 0.6002                       | 2.2092                        | 2.48105                      |
| Landau theory [80]                                |                                        | 0               | $1/2$                       | $1/2$    | $2/3$                        | $7/3$                         | -                            |
| Gaussian exponents [80]                           |                                        | $\frac{4-d}{2}$ | $\frac{d-2}{4}$             | $1/2$    | $\frac{4}{d+4}$              | $\frac{3d+4}{d+4}$            | $1 + \frac{d}{2}$            |

### 3. Key Attributes of Fractal Networks

The properties of fractal networks (see, for instance, Figure 2) can be characterized by a set of dimension number [81–84]. Formal definitions of key dimension numbers are replicated in Table 5. In particular, the scale invariance is usually characterized either by the self-similarity, or the box-counting dimension. The network connectedness can be classified via relations between the topological ( $d_t$ ), the connectivity ( $d_\ell$ ), and the topological connectivity ( $d_{t\ell}$ ) dimensions [84]. The network connectivity can be quantified by the averaged coordination number  $\langle z \rangle$  together with the connectivity dimension  $d_\ell$  [50]. The network ramification is characterized by the topological Hausdorff dimension [84–86]. Specifically, finitely ramified networks have  $D_{tH} = 1$ , whereas for infinitely ramified networks

$$D_{tH} = Q + 1 > 1 \quad (1)$$

where  $Q$  is the ramification exponent, sometimes called the connectivity [21].

**Table 5.** Basic dimension numbers characterizing the main features of fractal networks embedded in the Euclidean space  $E^d$  (see Refs. [81–84] and references therein).

| Dimension Number                               | Symbol      | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spatial dimension                              | $d$         | The maximum number of mutually orthogonal vectors in the embedding Euclidean space $E^d$ .                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Topological dimension                          | $d_t$       | $d_t(F) = \min \left\{ s : \text{there is a subset } X \subseteq F, \text{ such that } d_t(X) = s - 1, \right. \\ \left. \text{while } d(F \setminus X) \leq 0 \right\}$                                                                                                                                                                                                                                                                                                                                      |
| Hausdorff dimension                            | $D_H$       | $D_H = s$ , such that $\lim_{\varepsilon \rightarrow 0} H_\varepsilon^s(F) = \begin{cases} 0, & \text{if } s > D_H \\ \infty, & \text{if } s < D_H \end{cases}$ , where $H_\varepsilon^s(F) = \inf \left\{ \sum_{i=1}^{\infty}  U_i ^s : \{U_i\} \text{ is } \varepsilon\text{-cover of } F \right\}$ is the Hausdorff measure with respect to the Euclidean metric in $E^d$ , is a non-negative number and the infimum is taken over all countable d-dimensional balls with diameters $r \leq \varepsilon$ . |
| Topological Hausdorff dimension                | $D_{tH}$    | $D_{tH}(F) = \min \left\{ s : \text{there is a subset } X \subseteq F, \text{ such that } D_H(X) = s - 1, \right. \\ \left. \text{while } d_t(F \setminus X) \leq 0 \right\}$                                                                                                                                                                                                                                                                                                                                 |
| Connectivity dimension                         | $d_\ell$    | $d_\ell = \lim_{\varepsilon \rightarrow 0} [\ln N(\ell) / \ln \ell]$ , where $N(\ell)$ is the number of fractal points connected with an arbitrary point inside of the $d_\ell$ -ball of diameter $\ell$ .                                                                                                                                                                                                                                                                                                    |
| Topological connectivity dimension             | $d_{t\ell}$ | $d_{t\ell}(F) = \inf \{ s : \exists A \subseteq F \text{ such that } d_{H\ell}(A) \leq s - 1 \text{ and } d_t(F \setminus A) \leq 0 \}$ , where $d_{H\ell} = D_H(A) \times [d_\ell(F) / D_H(F)]$ is the Hausdorff dimension of the subset $A$ with respect to the geodesic metric on the network $F$ .                                                                                                                                                                                                        |
| Fractal dimension of the minimum path          | $d_{\min}$  | Fractal dimension of the minimum path is defined via the scaling relation $\langle \ell_{\min} \rangle \propto r^{d_{\min}}$ , where $\langle \dots \rangle$ denotes the ensemble average, while $\ell_{\min} \propto \inf \{ \ell(\gamma) : \gamma \text{ is a path joining } A \text{ to } B, A, B \in \text{CPC} \}$ and $r$ is the Euclidean distance between these points.                                                                                                                               |
| Random walk dimensions                         | $D_W, d_W$  | The random walk dimensions are defined via the scaling relation of the mean squared displacement of the random walker on the network $\langle r^2 \rangle \propto t^{D_W}$ and $\langle \ell^2 \rangle \propto t^{d_W}$ respectively, where $D_W = d_{\min} d_W$ .                                                                                                                                                                                                                                            |
| Spectral dimension                             | $d_s$       | The number of effective dynamical degrees of freedom of random walker in the network, which is equal to the number of propagating modes. Accordingly, $d_s = 2\partial \ln P(t) / \partial \ln t$ , where $P(t)$ is the propagator and $t$ is the diffusion time.                                                                                                                                                                                                                                             |
| Number of effective spatial degrees of freedom | $n_\gamma$  | The number of independent directions in the network in which a random walker can move without violating any constraint imposed on it by the network topology. In numerical simulations can be determined from the probability to find the walker at the distance $r$ from its origin after a fixed number of steps.                                                                                                                                                                                           |

The effective spatial and dynamical degrees of freedom on the self-similar network are controlled by the fractal loopiness [84]. The sum of numbers of effective spatial and dynamical degrees of freedom is equal to

$$n_\gamma + d_s = 2d_\ell, \quad (2)$$

while, generally,  $d_s \leq d_\ell$  and so  $n_\gamma \geq d_\ell$  [87]. Accordingly, the index of fractal loopiness is defined as [84].

$$\Lambda = d_s / n_\gamma - 1 / d_\ell, \quad (3)$$

The electrical conductivity of the fractal network is commonly characterized by the electrical resistance exponent is equal to

$$\zeta = D_W(1 - d_s/2), \quad (4)$$



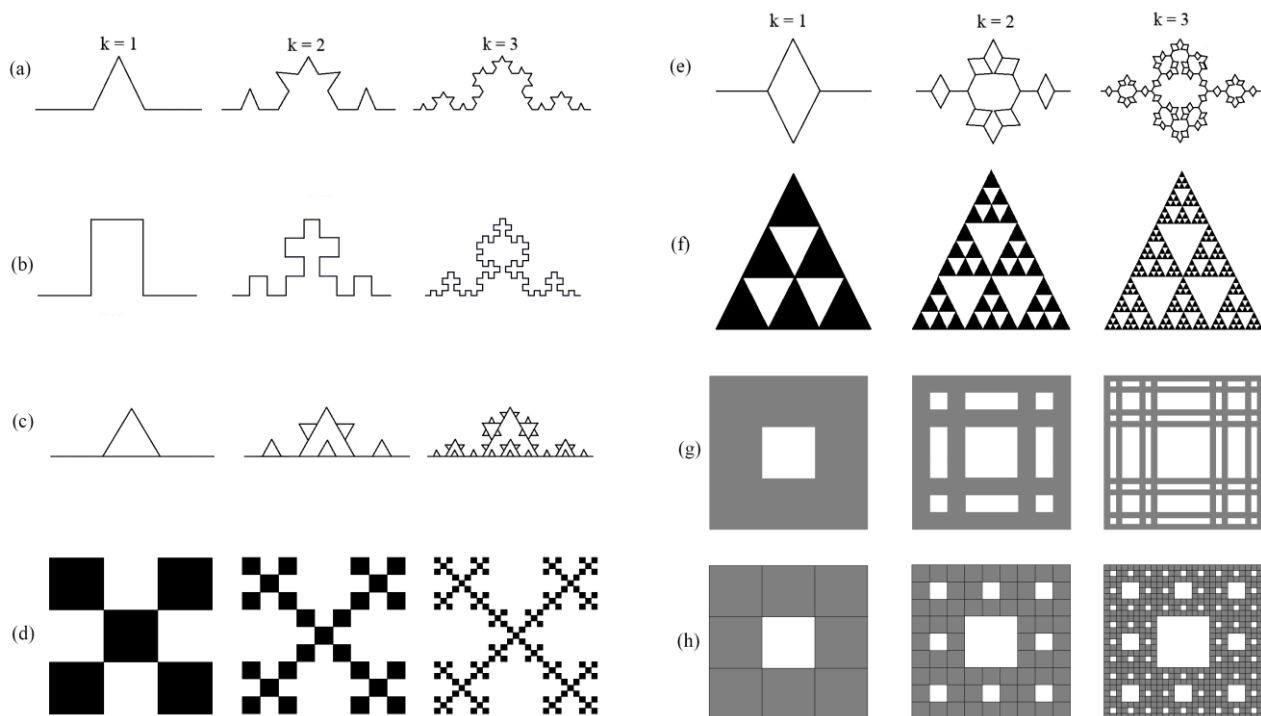
such that the loopless fractal networks are characterized by  $\Lambda = 0$ , whereas for networks with loops at all scales the fractal loopiness index is in the range of  $0 < \Lambda \leq 1 - 1/d_\ell < 1$  where is the fractal dimension of the random walk on the network [83]. Many fractals (e.g., all fractals shown in Figure 2) obey the Alexander–Orbach relation

$$d_s = 2D/D_W, \quad (5)$$

such that  $\zeta = D_W - D \leq d_{\min} = D/d_\ell$  [88]. For fractals obeying the Alexander–Orbach conjecture (5) the number of effective spatial degrees of freedom is equal to

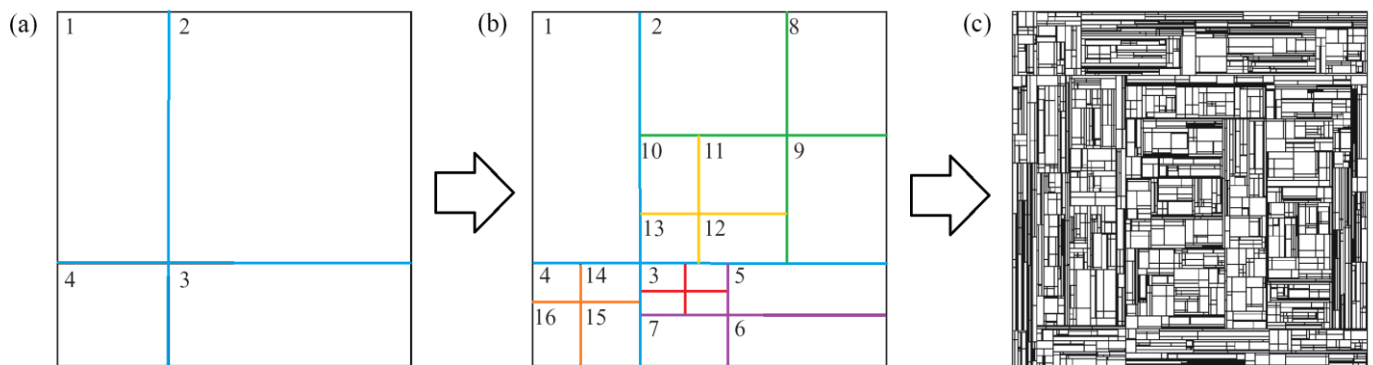
$$n_\gamma = d_s(d_W - 1), \quad (6)$$

such that  $n_\gamma$  can be less than, equal to, or greater than the fractal dimension  $D$ , while  $d_W = D_W/d_{\min}$  is the fractal dimension of the random walk with respect to the geodesic metric [87]. Thus, the fractal properties of self-similar network should be characterized by at least five dimension numbers. The values of the key dimension numbers for classical fractal networks shown in Figure 2 are given in Table 6. Notice that all fractals presented in Table 6 obey relation (5) and so only four reported dimension numbers are independent (e.g.,  $D_{tH}$ ,  $D_H$ ,  $d_{\min}$ , and  $d_s$ , or  $D_{tH}$ ,  $D_H$ ,  $d_s$ , and  $n_\gamma$ ). However, generally, four (or even five) dimension numbers may be insufficient to account for all fractal features of a self-similar network (see Ref. [84]).



**Figure 2.** Iterative construction of classical fractal networks. Three first iterations of: (a) Koch curve ( $D = \ln 4 / \ln 3$ ,  $d_{\min} = D$ ); (b) Koch curve ( $D = \ln 5 / \ln 3$ ,  $d_{\min} = D$ ); (c) branched Koch curve ( $D = \ln 5 / \ln 3 = d_\ell$ ,  $d_{\min} = 1$ ); (d) Vicsek snowflake ( $D = \ln 5 / \ln 3$ ,  $d_{\min} = 1$ ); (e) branched Koch curve ( $D = \ln 6 / \ln 3$ ,  $d_{\min} = \ln 4 / \ln 3$ ); (f) Sierpinski gasket ( $D = \ln 6 / \ln 3$ ,  $d_{\min} = 1$ ); (g) Cantor tartan ( $D = \ln 6 / \ln 3$ ,  $d_{\min} = 1$ ); (h) Sierpinski carpet ( $D = \ln 8 / \ln 3$ ,  $d_{\min} = 1$ ). Fractal attributes of these networks are summarized in Table 2, Refs. [5–10].





**Figure 3.** Iterative construction of weighted planar stochastic lattice (WPSL): (a) first iteration; (b) after 6 iterations; (c) snapshot of WPSL after 30 iterations.

**Table 6.** Dimension numbers for several fractal networks.

| Network                                      | $D_{tH}$                | $D_H$                              | $d_{\min}$      | $d_\ell$                            | $d_s$                                         | $n_\gamma$      | $\Lambda$                          |
|----------------------------------------------|-------------------------|------------------------------------|-----------------|-------------------------------------|-----------------------------------------------|-----------------|------------------------------------|
| Koch curve (Figure 2a)                       | 1                       | $\ln 4 / \ln 3$<br>$\approx 1.262$ | $\ln 4 / \ln 3$ | 1                                   | 1                                             | 1               | 0                                  |
| Koch curve (Figure 2b)                       | 1                       | $\ln 5 / \ln 3$<br>$\approx 1.465$ | $\ln 5 / \ln 3$ | 1                                   | 1                                             | 1               | 0                                  |
| Branched Koch curve (Figure 2c)              | 1                       | $\ln 5 / \ln 3$                    | 1               | $\ln 5 / \ln 3$                     | $2 \ln 5 / \ln(40/3)$<br>$\approx 1.243$ [89] | 1.597           | 0.152                              |
| Vicsek snowflake (Figure 2d)                 | 1                       | $\ln 5 / \ln 3$                    | 1               | $\ln 5 / \ln 3$                     | $\ln 25 / \ln 15$<br>$\approx 1.1886$         | 1.741           | 0                                  |
| Branched Koch curve (Figure 2e)              | 1                       | $\ln 6 / \ln 3$<br>$\approx 1.631$ | $\ln 4 / \ln 3$ | $\ln 6 / \ln 4$<br>$\approx 1.2925$ | -                                             | -               | -                                  |
| Sierpinski gasket (Figure 2f)                | 1                       | $\ln 6 / \ln 3$                    | 1               | $\ln 6 / \ln 3$                     | $2 \ln 6 / (90/7)$<br>$\approx 1.403$ [89]    | 1.8587          | 0.1418                             |
| Cantor tartan (Figure 2g)                    | $\ln 6 / \ln 3$<br>[90] | $\ln 6 / \ln 3$                    | 1               | $\ln 6 / \ln 3$                     | $\ln 6 / \ln 3$<br>[90]                       | $\ln 6 / \ln 3$ | $\ln 2 / \ln 6$<br>$\approx 0.387$ |
| Sierpinski carpet (Figure 2h)                | $\ln 6 / \ln 3$<br>[85] | $\ln 8 / \ln 3$<br>$\approx 1.893$ | 1               | $\ln 8 / \ln 3$                     | 1.8061<br>[85]                                | 1.9795          | 0.3841                             |
| WPSL (Figure 3) [40]                         | 2                       | 2                                  | $\geq 1$        | $\leq 2$                            | $\leq 2$                                      | $\leq 2$        | $\geq 0.5$                         |
| Critical percolation cluster in $d = 2$ [86] | 7/4                     | 91/48                              | 1.13077         | 1.6585                              | 1.3170                                        | 2               | 0.0555                             |
| Backbone of CPC in $d = 2$ [86]              | 1                       | 1.64336                            | 1.13077         | 1.45331                             | 1.256                                         | 1.65            | 0.073                              |
| Critical percolation cluster in $d = 6$ [83] | 3                       | 4                                  | 2               | 2                                   | 4/3                                           | 8/3             | 0                                  |
| Planar random trees [91]                     | 1                       | 2                                  | 1               | 2                                   | 4/3                                           | 8/3             | 0                                  |

### 3.1. Fractal Features of Infinitely Ramified Sierpinski Carpets

In particular, Sierpinski carpets constitute a generic model of infinitely fractal networks which was widely used to study physical phenomena in systems with a non-integer dimension. A standard infinitely ramified Sierpinski carpet  $S_n^m$  can be constructed as follows. The unit square  $[0, 1]^2$  is divided into  $n \times n$  sub-squares of equal size and  $m \times m$  sub-squares are deleted in the center of the initial square. This process is repeated on the remaining sub-squares ad infinitum. Figure 2h shows three steps of the iterative construction of standard Sierpinski carpet. The number of sites on the carpet  $N_k$  increases with number of interactions  $k$  as  $N_k = (n^2 - m^2)^k$ , while the carpet size (measured in

the number of sites) increases as  $L_k = n^k$ . Therefore, the self-similarity and Hausdorff dimensions of  $S_n^m$  both are equal to

$$D_H = \ln(n^2 - m^2) / \ln(n) < 2, \quad (7)$$

while the geodesic paths on the network are characterized by  $d_{\min} = 1$  and so, the connectivity dimension is equal to  $d_\ell = D_H$ . The topological Hausdorff dimension of standard Sierpinski carpet is equal to

$$D_{tH} = 1 + \ln(n - m) / \ln(n), \quad (8)$$

and so  $1 < d_\ell < D_H$  [85]. The arithmetically averaged coordination number of the pre-fractal Sierpinski carpet  $S_n^m(k)$  increases with number of iterations  $k$  as

$$\langle z_k \rangle = z_\infty \left[ 1 - n^{-(D_H-1)k} \right], \quad (9)$$

where

$$z_\infty = 4 \left( 1 - n^{1-D_H} \right) / \left( 1 - n^{1-D_H} \right) > 8/3, \quad (10)$$

is the averaged coordination number in the fractal limit of  $k \rightarrow \infty$  [50]. The constrictivity factor of the standard Sierpinski carpet is equal to

$$0 < \delta_c = 1 - m/n < 1, \quad (11)$$

independently of the number of iterations, such that  $\delta_c \rightarrow 0$  as  $m/n \rightarrow 1$  [92]. Although, there is no exact analytic expression for the spectral dimension of the infinitely ramified Sierpinski carpets, it was suggested the phenomenological relation

$$d_s = 2 \ln(n^2 - m^2) / \ln(n^2 + m^2), \quad (12)$$

which provides good approximation for standard Sierpinski carpets [85]. It was also recognized that the standard Sierpinski carpets obey the Alexander–Orbach conjecture (4) and so the main fractal attributes  $D_W$ ,  $\zeta$ ,  $n_\gamma$ , and  $\Lambda$  can be calculated using analytic relations.

On the other hand, it was also recognized that the fractal features of infinitely ramified Sierpinski carpets are also characterized by two other dimension numbers, which cannot be expressed in the term of the basic dimensions. Specifically, the carpet  $S_n^m(k)$  obtained after  $k$  iterations can be divided on  $n^k$  layers connected in series. Each layer can be characterized by the fractal dimension  $D_i^k = 2 - \ln[\phi_i(k)]$ , where  $\phi_i(k)$  is the layer porosity [92]. The mean fractal dimension of layers is found to be independent of number of iterations and is equal to

$$\langle D_i \rangle = 2 + (m/n) \ln(1 - m/n) / \ln(n). \quad (13)$$

Another dimension number characterizing the fractal properties of infinitely ramified Sierpinski carpets is the fractal dimension of the random walk on the carpet basis

$$D_{tW} = \ln(\delta_c n^2 + m^2) / \ln(n), \quad (14)$$

while the constrictivity factor  $\delta_c$  is defined by Equation (11) [92]. The values of adimensional factors and dimension numbers for some standard Sierpinski carpets are summarized in Table 7.

**Table 7.** Fractal attributes of standard Sierpinski carpets  $S_n^m$ .

| Carpet Parameters |     | Adimensional Factors |                    |                  | Basic Dimension Numbers |                       |               | Dimension Numbers             |                  |
|-------------------|-----|----------------------|--------------------|------------------|-------------------------|-----------------------|---------------|-------------------------------|------------------|
| $n$               | $m$ | $z_\infty$<br>(10b)  | $\delta_c$<br>(11) | $\Lambda$<br>(3) | $D_{tH}$<br>(9)         | $D_H = d_\ell$<br>(8) | $d_s$<br>(12) | $\langle D_i \rangle$<br>(13) | $D_{tW}$<br>(14) |
| 99                | 1   | 3.9995               | 98/99              | 0.499            | 1.9977                  | 1.9999                | 1.999         | 1.9999                        | 1.998            |
| 7                 | 1   | 3.9024               | 6/7                | 0.487            | 1.9207                  | 1.9894                | 1.979         | 1.9887                        | 1.933            |
| 5                 | 1   | 3.7894               | 4/5                | 0.47             | 1.8613                  | 1.9746                | 1.951         | 1.9723                        | 1.892            |
| 6                 | 2   | 3.6923               | 2/3                | 0.427            | 1.7737                  | 1.9342                | 1.879         | 1.9246                        | 1.86             |
| 7                 | 3   | 3.6363               | 4/7                | 0.393            | 1.7124                  | 1.8957                | 1.817         | 1.8767                        | 1.856            |
| 3                 | 1   | 3.2                  | 2/3                | 0.384            | 1.6309                  | 1.8927                | 1.806         | 1.8769                        | 1.771            |
| 4                 | 2   | 3                    | 1/2                | 0.303            | 1.5                     | 1.7924                | 1.659         | 1.75                          | 1.792            |
| 5                 | 3   | 2.9090               | 2/5                | 0.259            | 1.4306                  | 1.7227                | 1.572         | 1.6584                        | 1.829            |
| 6                 | 4   | 2.8571               | 1/3                | 0.231            | 1.3868                  | 1.6719                | 1.516         | 1.5912                        | 1.860            |
| 7                 | 5   | 2.8235               | 2/7                | 0.181            | 1.3562                  | 1.6331                | 1.445         | 1.5401                        | 1.883            |
| 99                | 97  | 2.6757               | 2/99               | 0.103            | 1.1508                  | 1.2994                | 1.211         | 1.168                         | 1.998            |

### 3.2. Cantor Tartans

Cantor tartans (see, for example, Figure 2g) represent a special class of infinitely ramified networks with unique fractal attributes [90]. The Cantor tartan  $CT_n^m$  is a subset of the Sierpinski carpet  $S_n^m$ , such that both have the same topological Hausdorff dimension

$$D_{tH}(CT_n^m) = D_{tH}(S_n^m) = 1 + \ln(n - m) / \ln(n), \quad (15)$$

while the Hausdorff dimension of the Cantor tartan is equal to its topological Hausdorff dimension  $D_H(CT_n^m) = D_{tH}(CT_n^m)$ . Furthermore, the mean fractal dimension of layers and the spectral dimension are both equal to the Hausdorff dimension. Consequently,

$$d_s = d_\ell = \langle D_i \rangle = D_{tH} = D_H, \quad (16)$$

such that

$$D_W = d_W = D_{tW} = 2, \quad (17)$$

and so, the fractal loopiness index is equal to  $\Lambda = 1 - 1/D_H$ . Therefore, the basic fractal attributes of the Cantor tartan are determined by the unique dimension number, e.g.,  $D_H$ .

### 3.3. Multifractal Weighted Planar Stochastic Lattice

Another model used to study the effect of fractal features on percolation is a weighted planar stochastic lattice (WPSL) formed by the random sequential partition of a plane into contiguous and non-overlapping blocks [39–41]. Schematic illustration of the first few steps of iterative construction of the WPSL is shown in Figure 3. At the first step, the generator randomly divides a square into four smaller blocks. These blocks are labeled by their respective areas in a clockwise fashion, starting from the upper left block. In each step thereafter, only one block is picked preferentially with respect to the respective and it is then divided randomly into four blocks. The steps repeated ad infinitum. Although the coordination number of each block in the constructed network is random, it has been established that the degree distribution decays obeying power law:

$$P(k) \propto k^{-\theta}, \quad (18)$$

where the scaling exponent is found to be equal to  $\theta = 5.66$  and the mean coordination number is  $\langle k \rangle = \sum_k kP(k) = 5.333$  [40], that is greater than the coordination number of the

square network, the deterministic counterpart of the WPSL. The scaling properties of the WPSL are characterized by the spectrum of generalized dimensions Renyi  $D_q \leq 2$ , while the box-counting dimension is equal to  $D_H = D_{q=0} = 2$  [39]. The key dimension numbers of the WPSL are summarized in Table 6.

#### 4. Percolation on Finitely Ramified Fractal Networks

For critical phenomena, the network ramification plays a more imperative role than periodicity and scale invariance. The fractal networks having a finite order of ramification can be considered ‘marginal’ between one-dimensional and higher-dimensional geometries. Fortunately, physical models defined on finitely ramified networks are exactly solvable [19–24]. In particular, it has been recognized that the finitely ramified networks have only a trivial threshold,  $p_C = 1$ , for both site and bond percolation problems. Indeed, this is obvious for self-avoiding fractal chains, e.g., Koch curves in Figure 2a,b. For percolation on Sierpinski gaskets and branching Koch curves the value  $p_C = 1$  was obtained as the limit of apparent percolation thresholds on pre-fractal networks  $p_c(k) \rightarrow 1$  as the number of network iterations  $k \rightarrow \infty$ . Specifically, with the help of exact renormalization group recursion relations it was found that

$$p_c \propto 1 - 0.5k^{-1/2} \quad (k \gg 1), \quad (19)$$

for the bond percolation [20] and

$$p_c \propto 1 - k^{-1} \quad (k \gg 1), \quad (20)$$

for the site percolation on Sierpinski gasket (Figure 2f) [22]. For the bond percolation on the branching Koch curve (Figure 2c) it was found that

$$p_c \propto 1 - 2^{-k}, \quad (21)$$

while  $k \gg 1$  [23]. These findings were also confirmed by numerical simulations [22,23].

Critical exponents for percolation on branching Koch curves were obtained in Ref. [24]. It has been also established that for percolation on fractal networks the critical exponents obey the scaling relations

$$2 - \alpha = \gamma + 2\beta = D\nu + \beta = D_H\nu, \quad (22)$$

which resemble the scaling relations for percolation on the Euclidean networks (see Table 5), but the topological dimension  $d$  is replaced by the fractal dimension of the network  $D_H$  [24]. In this regard, it is easy to understand that the condition  $p_C = 1$  implies that the fractal dimension of the critical percolation cluster on the finitely ramified network is equal to the network fractal dimension, that is,

$$D = D_H. \quad (23)$$

From Equations (22) and (23), it immediately follows that the percolation on a finitely ramified network is characterized by  $\beta = 0$ , while the number of iterations  $k \rightarrow \infty$ . Furthermore, taking into account the scaling relation  $\tau = 1 + (2 - \alpha)/(2 - \alpha - \beta)$ , we expect that the cluster size distribution on the finitely ramified network is characterized by the universal exponent  $\tau = 2$ , independently of the network fractal dimension. Accordingly, the universality classes of the Bernoulli percolation on the finitely ramified networks are determined by the self-similarity dimension  $D_H$  and the index of fractal loopiness (3), while other dimension numbers characterizing the finitely ramified networks are univalued functions of  $D_H$  and  $\Lambda$ .

#### 5. Percolation on Infinitely Ramified Fractal Networks

In contrast to finitely ramified fractals, the infinitely ramified fractal networks allows for interpolate regular lattices in non-integer dimensions [93]. Accordingly, the critical

exponents for Ising-like systems in non-integer dimensions were studied by different methods for  $1 < D \leq 4$  [94–100].

Against this background, Shinoda [46] has proved the existence of phase transition of Bernoulli percolation on the Sierpinski carpet lattice. Higuchi and Wu [48] have established that the critical probability for the Sierpinski carpet network in two dimensions is uniquely determined and the percolation transition is sharp. Monceau and Hsiao [47] have performed Monte Carlo simulations of the Bernoulli site percolation on pre-fractal Sierpinski carpets. They noted that the scaling corrections occurring in the behavior of the thresholds with the size of the network are stronger than for the percolation on translationally invariant lattices. The percolation behavior on the infinitely ramified Sierpinski carpets is found to obey the scaling relations (22) which were firstly established for percolation on the finitely ramified fractal networks. However, in contrast to Equation (23), the fractal dimension of the critical percolation cluster is always less than the fractal dimension of the Sierpinski carpet network and so the critical exponent  $\beta$  is always positive. Comprehensive simulations of the Bernoulli site percolation on infinitely ramified Sierpinski networks and Cantor tartans were performed in Refs. [49,50]. The Bernoulli site and bond percolation on infinitely ramified multifractal networks was studied in Refs. [38–42].

### 5.1. Percolation on Sierpinski Carpets

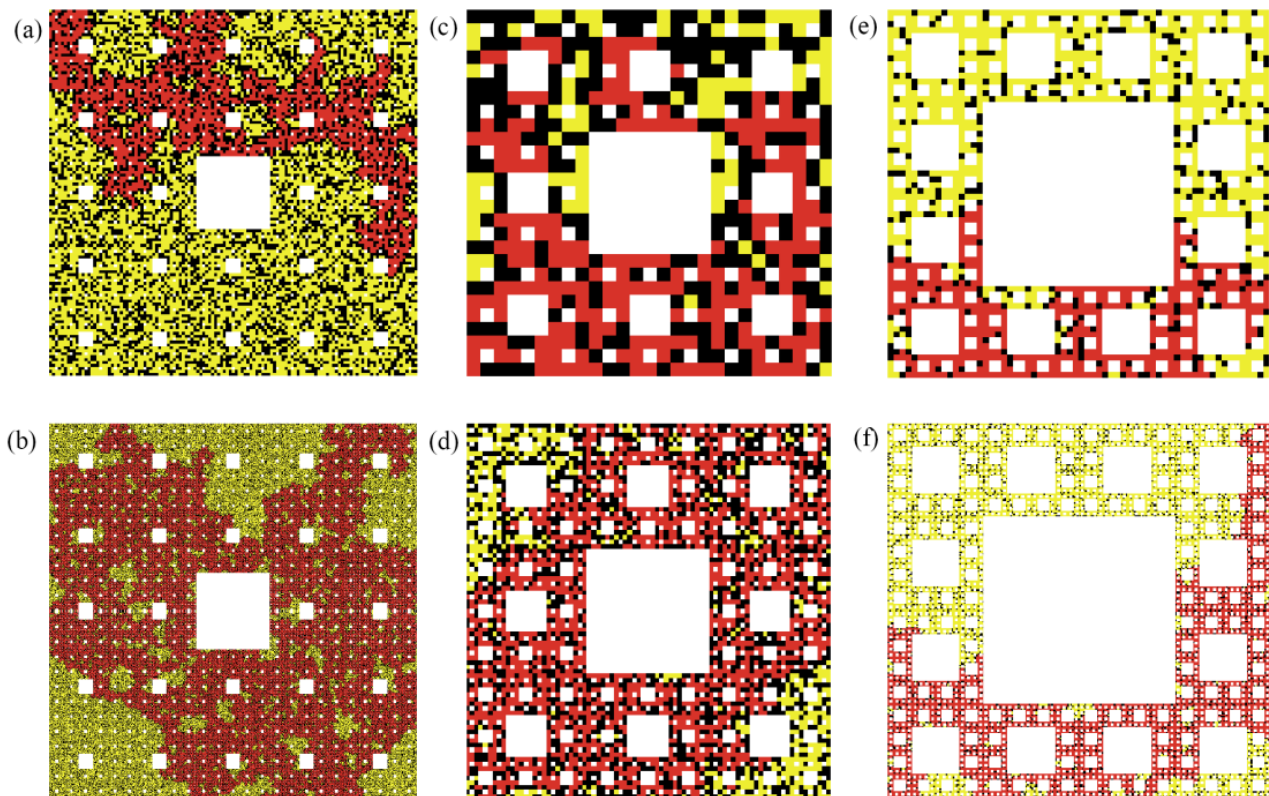
In Refs. [47,49,50] Monte Carlo simulations were performed on pre-fractal Sierpinski carpets  $S_n^m$  with free boundaries. The network sites were occupied one by one in random order. Occupied sites form contiguous clusters which were identified using the breadth-first search algorithm (see Ref. [101]). A percolation cluster was defined as the cluster that spans the carpet of size  $L_k = n^k$  across either the vertical or horizontal directions, or both (see, for illustration, Figure 4). Accordingly, there are two types of spanning probability defined as the likelihood of finding the spanning cluster. In practice, the spanning probability is commonly defined as  $P_k(p) = 0.5[P_k^x(p) + P_k^y(p)]$ , where  $P_k^y(p)$  and  $P_k^x(p)$  are the probabilities that the spanning cluster span the carpet along the vertical and horizontal directions, respectively. The graphs of  $P_k(p)$  for some Sierpinski carpets are shown in Figure 5a–d. For pre-fractal networks, there is a small transition region in which the spanning probability increases from zero towards one. In the limit of  $k \rightarrow \infty$  the spanning probability is expected to be a step function. For regular lattices, the renormalization theory defines the fixed point  $P_L(p_C) = p_C$ , such that  $|p_c(L) - p_C| \propto L^{-1/\nu}$  [102]. Although the equality  $P_L(p_C) = p_C$  fails for fractal networks, the apparent threshold can be defined via condition  $P_k(p_a) = C$  for any  $0 < C < 1$ , such that  $p_a(C, k \rightarrow \infty) \rightarrow p_C$  [49]. Empirically, it was found that the apparent critical threshold for the pre-fractal Sierpinski carpets depends on the number of iterations as

$$p_a(k, C) = p_C + c_1 n^{-k/\nu} + c_2 n^{-2k/\nu} + c_3 n^{-k\omega}, \quad (24)$$

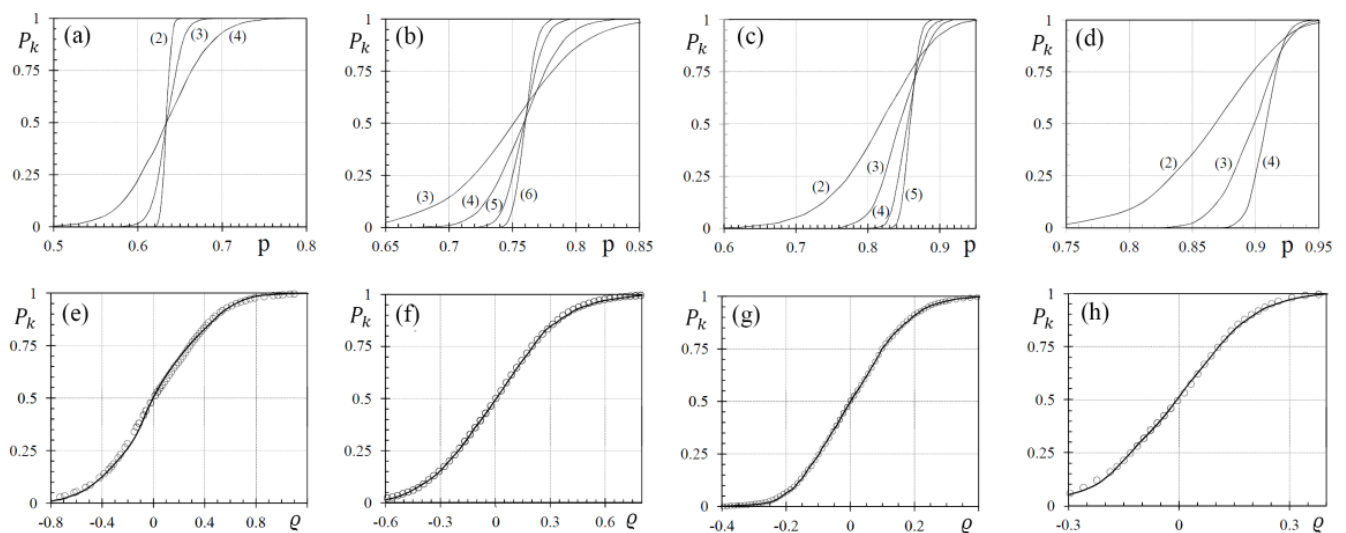
where  $c_i$  are the fitting constants and  $\omega > 2/\nu$  [49]. Accordingly, for sufficiently large  $k$  the spanning probabilities can be collapsed into a single curve in coordinates  $P_k$  versus  $q = [p - p_a(k, C)]n^{k/\nu}$ , as is shown in Figure 5e–h. The values of site percolation thresholds (24) and critical exponents obtained in the numerical simulations on pre-fractal Sierpinski carpets in [47–50] are summarized in Table 8, Refs. [49,50]. It was recognized that the critical exponents obey the standard scaling relation  $\alpha + 2\beta + \gamma = 2$ , as well as the generalized hyperscaling relations

$$\gamma = \nu D_H - 2\beta \text{ and } D = D_H - \beta/\nu, \quad (25)$$

whereas the hyperscaling relation  $\tau = D_H/D$  was not firmly confirmed. Nonetheless, it has been suggested that the universality classes of the Bernoulli percolation on the standard Sierpinski carpets are controlled by three dimension numbers, e.g.,  $D_{tH}$ ,  $D_H$  and  $d_s$  [49].



**Figure 4.** Percolation on pre-fractal Sierpinski carpets: (a,b)  $S_5^1$ ; (c,d)  $S_3^1$ ; (e,f)  $S_4^2$  after three (a,c,e) and four (b,d,f) iterations. Percolation clusters—red; occupied sites—yellow; empty sites—black.



**Figure 5.** Spanning probabilities on the pre-fractal Sierpinski carpets. The spanning probabilities versus the occupation probability for the pre-fractal Sierpinski carpets: (a)  $S_5^1$ ; (b)  $S_3^1$ ; (c)  $S_4^2$ ; (d)  $S_5^3$ . Numbers in parentheses indicate the number of iterations of the pre-fractal carpet. Collapse curves in coordinates  $P_k$  versus  $q = [p - p_a(k, C)]n^{k/v}$  for spanning probabilities on the pre-fractal Sierpinski carpets: (e)  $S_5^1$ ; (f)  $S_3^1$ ; (g)  $S_4^2$ ; (h)  $S_5^3$  (threshold values  $p_a$  are defined by the relation  $P_k(p_a) = 0.5$ ).

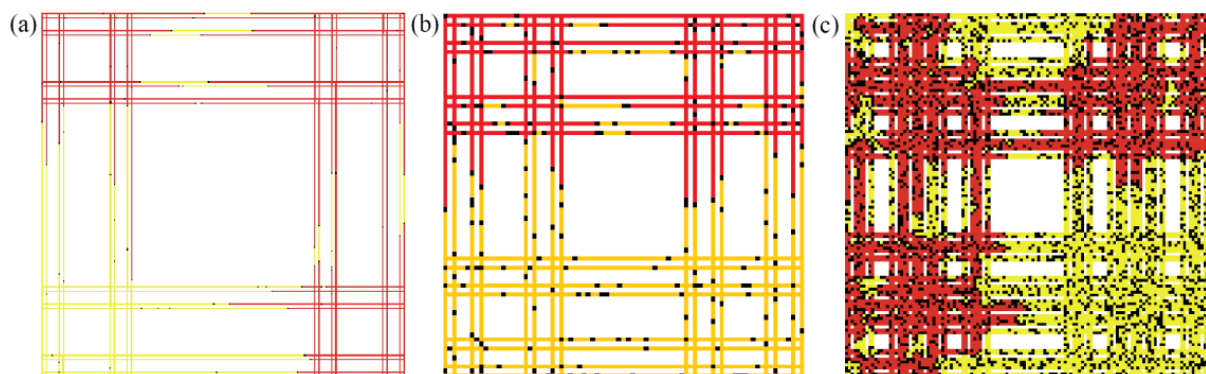


**Table 8.** Site percolation thresholds and critical exponents for the Bernoulli site percolation on the standard Sierpinski carpets and Cantor tartans.

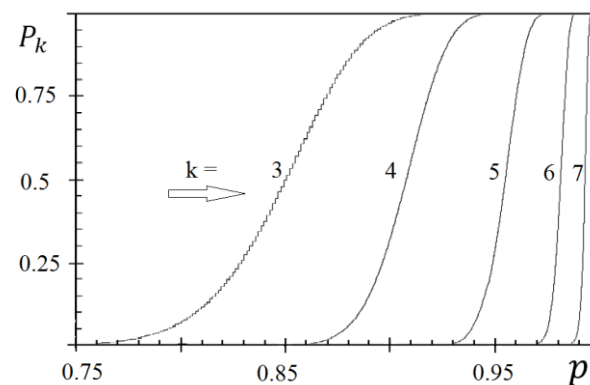
|          | $S_5^1$           | $S_3^1$           | $S_4^2$           | $S_5^3$           | $CT_5^1$          | $CT_3^1$        |
|----------|-------------------|-------------------|-------------------|-------------------|-------------------|-----------------|
| $D_H$    | 1.9746            | 1.8927            | 1.7924            | 1.7227            | 1.8613            | 1.6309          |
| $p_C$    | $0.634 \pm 0.004$ | $0.758 \pm 0.002$ | $0.858 \pm 0.004$ | $0.914 \pm 0.004$ | $\geq 0.725$      | $1 \pm 0.002$   |
| $D$      | 1.885             | 1.829             | 1.767             | 1.718             | 1.855             | $1.631 = D_H$   |
| $\alpha$ | −0.97             | −1.40             | −2.75             | −2.10             | −0.69             | −               |
| $\beta$  | $0.134 \pm 0.004$ | $0.114 \pm 0.004$ | $0.067 \pm 0.003$ | $0.011 \pm 0.003$ | $0.010 \pm 0.007$ | 0               |
| $\nu$    | $1.50 \pm 0.01$   | $1.790 \pm 0.007$ | $2.65 \pm 0.04$   | $2.40 \pm 0.05$   | $1.45 \pm 0.05$   | −               |
| $\gamma$ | 2.69              | 3.16              | 4.6               | 4.1               | 2.66              | −               |
| $\tau$   | $2.05 \pm 0.05$   | $2.03 \pm 0.05$   | $2.01 \pm 0.04$   | $2.00 \pm 0.02$   | $2.02 \pm 0.05$   | $2.00 \pm 0.05$ |

### 5.2. Percolation on Cantor Tartans

The Bernoulli site percolation on Cantor tartans was studied in Ref. [50]. Monte Carlo simulations were performed on pre-fractal Cantor tartan networks with free boundaries. Typical configurations at the percolation threshold on Cantor tartans are shown in Figure 6. The percolation thresholds were determined from the data fittings with Equation (24). It was found that the site percolation threshold increases with increase the number of iterations (see, for instance, Figure 7), because the averaged coordination number of the Cantor tartan network  $\langle z_k \rangle$  decreases as  $k$  increases. Furthermore, it has been recognized that Cantor tartans  $CT_n^{n-2}$  are characterized by trivial site percolation threshold  $p_C = 1$  independently of  $D_H$ , because in the limit of  $k \rightarrow \infty$  the coordination number of  $CT_n^{n-2}$  is equal to  $\langle z_\infty \rangle = 2$  [50]. It was also found that the spanning cluster occupies about 50% of sites in  $CT_n^{n-2}$ , such that its mass fractal dimension is equal to the network fractal dimension, that is  $D = D_H$ , and  $\beta = 0$  and  $\tau = 2$  independently of  $D_H$ . Conversely, for the Cantor tartan  $CT_5^1$  with  $\langle z_\infty \rangle = 3$ , it was established that  $0 < p_C < 1$  (see Table 8, Refs. [49,50]). In this regard, it was also argued that other Cantor tartan with  $z_\infty > 2$  also have non-trivial site percolation thresholds [50]. The critical exponents for the Bernoulli site percolation on  $CT_3^1$  and  $CT_5^1$  are listed in Table 8, Refs. [49,50]. It is expected that the class of universality of the Bernoulli percolation on the Cantor tartan is determined by unique dimension number, e.g., the Hausdorff or self-similarity dimension [50].

**Figure 6.** Percolation on pre-fractal Cantor tartans after four iterations: (a)  $CT_5^3$ ; (b)  $CT_3^1$ ; (c)  $CT_5^1$ . Percolation clusters—red; occupied sites—yellow; empty sites—black.





**Figure 7.** Spanning probability versus occupation probability for pre-fractal Cantor tartans  $CT_3^1$  (numbers denote the number of iteration steps in the network construction).

### 5.3. Percolation on Critical Percolation Clusters and on WPSL

Critical percolation clusters formed on regular lattices constitute a special class of statistically self-similar networks [12,13]. Although, per definition, a backbone of the critical percolation cluster (CPC) is finitely ramified, the CPC is infinitely ramified, because it contains an infinite number of backbones between different pairs of points [86]. The bond percolation on critical percolation clusters was studied in [45]. Furthermore, the authors of [45] have introduced a lattice model in which percolation is constructed on top of critical percolation clusters recursively. The existence of non-trivial recursive percolation was observed in two and three dimensions. In two dimensions, the percolation thresholds were established up to  $k = 4$  generation. Moreover, the authors of [45] proposed an approximate empirical relation for the bond percolation thresholds on CPC of different generation on the square lattice  $p_C(k) = (k + 1)/(k + 2)$ . Notice that  $p_C(k = 0) = 0.5$  corresponds to the bond percolation on the square lattice, whereas for the bond percolation on the CPC of the first generation it was found  $p_C(k = 1) = 0.6549$ . It was also found that the fractal dimension of the red bonds  $d_R$  on the CPC quickly decreases with the number of generations. For percolation clusters  $d_R = D_{tH} - 1$  [86], and so the decrease of  $d_R$  indicates the decrease of ramification, that is consistent with the decrease of  $p_C(k)$ . On the other hand, it was found that the fractal dimension of the CPC only slightly decreases as  $k$  increases, whereas the fractal dimension of the CPC backbone increases with  $k$ . Accordingly, the authors of [45] concluded that the recursive percolation constitute a new universality class. However, to our best knowledge, there were not more works devoted to the recursive percolation.

Hassan and Rahman [39–41] studied site and bond percolation on a weighted planar stochastic lattice (WPSL), which is multifractal and whose dual is a scale-free network. The site and bond percolation thresholds are reported in Table 4. Although the Hausdorff dimension of the WPSL is equal to  $D_H = d = 2$ , it was found that the Bernoulli percolation on WPSL belong to a separate universality class than on all other planar lattices. The values of critical exponents for the Bernoulli percolation on the WPSL are presented in Table 4.

### 5.4. Effects of Network Connectivity and Ramification on Site Percolation Threshold

The knowledge of percolation thresholds is of paramount significance for applications of percolation in many fields of science and engineering [103–115]. In this regard, it was argued that a relevant dimension for the site percolation threshold on the fractal network is the topological Hausdorff dimension, rather than the topological or spectral dimensions [50]. From numerical simulations, it was found that the site percolation thresholds on the two-dimensional regular lattices and fractal networks can be well fitted by the following empirical formula

$$p_C = 1 / \left\{ 1 + c(D_{tH} - 1) \ln \left[ (D_{tH} - 1)^{0.5} (z_\infty - 1) \right] \right\} \quad (26a)$$

with the fitting constant  $c = 1/\sqrt{5} = 0.62133 \dots$ , while

$$(D_{tH} - 1)^{0.5}(z_{\infty} - 1) \geq 1. \quad (26b)$$

Notice that the requirement fails for the Cantor tartans  $CT_n^{n-2}$  with  $n \geq 3$ , as well as for the standard Sierpinski carpets constructed with  $n > 10$  and  $n - m \ll n$ , e.g., for  $S_{11}^9$  having  $z_{\infty} = 2.7586$  and  $D_{tH} = 1 + \ln 2 / \ln 11 = 1.2890$ . Furthermore, there are many translation invariant lattices which do not obey Equation (26a), even while condition (26b) holds (see Ref. [50]). So, generally, the knowledge of  $D_{tH}$  and  $z_{\infty}$  is insufficient for calculate the percolation threshold for infinitely ramified network. The answer to question of which attributes define the site and bond percolation thresholds remains a challenging problem.

### 5.5. About Universality Classes of Site Percolation on Fractal Networks

The percolation transition is characterized by a set of critical exponents which describe how the properties of the model are dependent on the observation scale. The critical exponents are often assumed to be functions of the system dimension, such that apparently unrelated models are often found to share the same critical exponents. This phenomenon is known as universality, and systems with identical exponents can be grouped together into universality classes [15,74,80]. Underlying the paradigm of universality is the fact that a translation invariant lattice has a unique well-defined dimension number  $d$  which determines all of its large-scale geometric features via their common scaling limit. However, as we pointed out in Section 3, different geometric and topological features of fractal networks are associated with different dimension numbers. This raises several challenging questions: (1) Which features are relevant for critical behavior? [17–21,49]. (2) Is it possible that the critical behavior depends only on a finite set of dimension numbers? [49–51]. (3) To what extent do the answers to these questions depend on the model under consideration? [9,36,51]. (4) How universal is universality? [51].

In Ref. [50], these questions were studied in two classes of geometric setting: transitive (including non-Euclidean) lattices with polynomial volume growth and self-similar fractals. Obtained results in these two cases push in opposite directions. Specifically, it was found that for transitive lattices, the critical exponents depend only on the topological dimension, suggesting that a very strong form of universality should hold in this setting. On the other hand, there were constructed two self-similar fractals for which a large number of standard dimensions coincide but which do not appear to have the same critical exponents for Bernoulli bond percolation. Numerical findings reported in Ref. [51] are summarized in Table 9. The authors of [51] argue that these data provide the evidence against the conjecture raised in Ref. [49] that the universality class on a self-similar network can be determined by an appropriate set of fractal attributes. Moreover, the authors of [51] suggest that no strong universality should be expected to hold for critical phenomena on self-similar fractals.

In this regard, first of all, we stress that for the Bernoulli percolation on the standard Sierpinski carpets, the critical exponents are governed by three dimension numbers [49]. For the Bernoulli percolation on the standard Cantor tartans, the set of fractal attributes governing the universality class of Bernoulli percolation is reduced to two numbers [50]. More generally, the characteristic features of infinitely ramified Sierpinski carpets should be characterized by more fractal parameters [84], which can affect the percolation behavior. Nonetheless, we expect that the universality class of the Bernoulli percolation on any infinitely ramified Sierpinski carpet can be determined by an appropriate set of fractal attributes, probably containing more than three dimension numbers.

In regard to the findings reported in Table 9, we note that although networks  $H_1$  and  $H_2$  are characterized by the same values of four dimension numbers reported in Table 9, the geometric features of these networks are clearly different.

**Table 9.** Dimension numbers characterizing the fractal properties of self-similar networks studied in [51] and percolation parameters on these networks. Notice that the self-similar trees  $T_i$  are finitely ramified and so they have only the trivial percolation threshold  $p_C = 1$ , whereas the networks obtained as the Cartesian products are infinitely ramified.

| Self-Similar Network                                    | $d$ | $D_{tH}$ | $D_H$ | $d_s$ | $p_C$   | $\sigma$ | $\tau$ |
|---------------------------------------------------------|-----|----------|-------|-------|---------|----------|--------|
| $T_O$                                                   | 1   | 1        | 2     | 4/3   | 1       | -        | -      |
| $T_I$                                                   | 1   | 1        | 2     | 4/3   | 1       | -        | -      |
| $T_{3/2}$                                               | 1   | 1        | 3/2   | 6/5   | 1       | -        | -      |
| $T_3$                                                   | 1   | 1        | 3     | 3/2   | 1       | -        | -      |
| $T_4$                                                   | 1   | 1        | 4     | 8/5   | 1       | -        | -      |
| $H_1 = T_O \times [0, 1]$                               | 2   | 3        | 3     | 7/3   | 0.4249  | -        | 2.195  |
| $H_2 = T_I \times [0, 1]$                               | 2   | 3        | 3     | 7/3   | 0.4232  | 0.385    | 2.151  |
| $H_3 = T_{3/2} \times T_{3/2} \times T_4 \times [0, 1]$ | 4   | 8        | 8     | 5     | 0.11705 | 0.41     | 2.66   |
| $H_4 = T_3 \times T_3 \times [0, 1] \times [0, 1]$      | 4   | 8        | 8     | 5     | 0.11326 | 0.41     | -      |

## 6. Conclusions

In this work we survey the analytical and numerical studies of percolation on fractal networks reported in the literature. We note that most of numerical simulations were performed on deterministic pre-fractal networks in two-dimensions. In this way, it was recognized that the percolation threshold is mainly controlled by the network ramification which is characterized by the topological Hausdorff dimension  $D_{tH}$ . Specifically, it has been established that finitely ramified networks ( $D_{tH} = 1$ ) have only a trivial threshold  $p_C = 1$  for the site and bond percolation. For infinite networks, the percolation threshold depends also on the network loopiness, connectivity, and symmetry. In this survey, we highlighted approximate relations accounting for the network connectivity and ramification. Nonetheless, we point out that the explicit analytical expression for percolation thresholds in terms of network attributes remains a challenging topic.

In this background, we argue that the universality class of Bernoulli percolation can be determined by a suitable set of dimension numbers characterizing the fractal features of the network. In particular, in the case of standard Sierpinski carpets and Cantor tartans the universality class can be determined by three independent dimension numbers, e.g., the topological Hausdorff, the connectivity, and the spectral dimensions. However, there is evidence that even four dimension numbers are insufficient to determine the class of universality for some types of fractal networks. Nevertheless, we expect that possible changes in process universality can be accounted for with the help of additional dimension numbers. However, the proof of this conjecture remains a challenging task. Another challenging problem is to understand the percolation on the stochastic fractals and, in particular, the recursive percolation.

Summarizing, we stress that the effects of fractal features on the Bernoulli percolation is far from understood. So, we expect that our survey will stimulate further research in this area.

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## References

- Kirkpatrick, S. Percolation and Conduction. *Rev. Mod. Phys.* **1973**, *45*, 574–588. [\[CrossRef\]](#)
- Stauffer, D.; Coniglio, A.; Adam, M. Gelation and critical phenomena. *Polym. Netw. Adv. Polym. Sci.* **1982**, *44*, 103–158. [\[CrossRef\]](#)
- Li, M.; Liu, R.R.; Lü, L.; Hua, M.B.; Xu, S.; Zhang, Y.C. Percolation on complex networks: Theory and application. *Phys. Rep.* **2021**, *907*, 1–68. [\[CrossRef\]](#)
- Broadbent, S.R.; Hammersley, J.M. Percolation processes. I. Crystals and mazes. *Proc. Camb. Philos. Soc.* **1957**, *53*, 629–641. [\[CrossRef\]](#)
- Sahimi, M. *Applications of Percolation Theory*; Taylor and Francis: London, UK, 1994.
- Araujo, N.; Grassberger, P.; Kahng, B.; Schrenk, K.J.; Ziff, R.M. Recent advances and open challenges in percolation. *Eur. Phys. J. Spec. Top.* **2014**, *223*, 2307–2321. [\[CrossRef\]](#)
- Saberi, A.A. Recent advances in percolation theory and its applications. *Phys. Rep.* **2015**, *578*, 1–32. [\[CrossRef\]](#)
- Stauffer, D.; Aharony, A. *Introduction to Percolation Theory*, 2nd ed.; Taylor and Francis: London, UK, 1994. [\[CrossRef\]](#)
- Dorogovtsev, S.N.; Goltsev, A.V.; Mendes, J.F.F. Critical phenomena in complex networks. *Rev. Mod. Phys.* **2008**, *80*, 1275–1335. [\[CrossRef\]](#)
- Stauffer, D. Scaling theory of percolation clusters. *Phys. Rep.* **1979**, *54*, 1–74. [\[CrossRef\]](#)
- Essam, J.W. Percolation theory. *Rep. Prog. Phys.* **1980**, *43*, 833–912. [\[CrossRef\]](#)
- Sokolov, I.M. Dimensionalities and other geometric critical exponents in percolation theory. *Sov. Phys. Usp.* **1986**, *29*, 924–945. [\[CrossRef\]](#)
- Isichenko, M.B. Percolation, statistical topography, and transport in random media. *Rev. Mod. Phys.* **1992**, *64*, 961–1043. [\[CrossRef\]](#)
- Wu, F.Y. The Potts model. *Rev. Mod. Phys.* **1982**, *54*, 235–268. [\[CrossRef\]](#)
- Ódor, G. Universality classes in nonequilibrium lattice systems. *Rev. Mod. Phys.* **2004**, *76*, 663–724. [\[CrossRef\]](#)
- Fortuin, C.M.; Kasteleyn, P.W. On the random-cluster model: I. Introduction and relation to other models. *Physica* **1972**, *57*, 536–564. [\[CrossRef\]](#)
- Gefen, Y.; Mandelbrot, B.B.; Aharony, A. Critical Phenomena on Fractal Lattices. *Phys. Rev. Lett.* **1980**, *45*, 855–858. [\[CrossRef\]](#)
- Suzuki, M. Phase Transition and Fractals. *Prog. Theor. Phys.* **1983**, *69*, 65–75. [\[CrossRef\]](#)
- Gefen, Y.; Aharony, A.; Mandelbrot, B.B. Phase transitions on fractals. I. Quasi-linear lattices. *Phys. A Math. Gen.* **1983**, *16*, 1267–1278. [\[CrossRef\]](#)
- Gefen, Y.; Aharony, A.; Shapir, Y.; Mandelbrot, B.B. Phase transitions on fractals. II. Sierpinski gaskets. *Phys. A Math. Gen.* **1984**, *17*, 435–444. [\[CrossRef\]](#)
- Gefen, Y.; Aharony, A.; Mandelbrot, B.B. Phase transitions on fractals. III. Infinitely ramified lattices. *Phys. A Math. Gen.* **1984**, *17*, 1277–1289. [\[CrossRef\]](#)
- Bo-Ming, Y.; Kai-Lun, Y. Numerical evidence of the critical percolation probability  $P_c = 1$  for site problems on Sierpinski gaskets. *J. Phys. A Math. Gen.* **1988**, *21*, 3269–3274. [\[CrossRef\]](#)
- Taitelbaum, H.; Havlini, S.; Grassberger, P.; Moenig, U. Percolation thresholds on finitely ramified fractals. *J. Phys. A Math. Gen.* **1990**, *23*, 371–373. [\[CrossRef\]](#)
- Wu, S.; Yang, Z.R. Bond percolation on branching Koch curves. *J. Phys. A Math. Gen.* **1995**, *28*, 2729–2735. [\[CrossRef\]](#)
- Trugman, S.A.; Weinrib, A. Percolation with a threshold at zero: A new universality class. *Phys. Rev. B* **1985**, *31*, 2974–2980. [\[CrossRef\]](#) [\[PubMed\]](#)
- Weinrib, A. Percolation threshold of a two-dimensional continuum system. *Phys. Rev. B* **1982**, *26*, 1352–1361. [\[CrossRef\]](#)
- Serrano, M.A.; Krioukov, D.; Boguna, M. Percolation in Self-Similar Networks. *Phys. Rev. Lett.* **2011**, *106*, 048701. [\[CrossRef\]](#)
- Rozenfeld, H.D.; ben-Avraham, D. Percolation in hierarchical scale-free nets. *Phys. Rev. E* **2007**, *75*, 056115. [\[CrossRef\]](#) [\[PubMed\]](#)
- Havlin, S.; Ben-Avraham, D.; Movshovitz, D. Percolation on Fractal Lattices. *Phys. Rev. Lett.* **1983**, *51*, 2347–2350. [\[CrossRef\]](#)
- Ben-Avraham, D.; Havlin, S.; Movshovitz, D. Infinitely ramified fractal lattices and percolation. *Phil. Mag. B* **1984**, *50*, 297–306. [\[CrossRef\]](#)
- Havlin, S.; Ben-Avraham, D.; Movshovitz, D. Percolation on Infinitely Ramified Fractals. *J. Stat. Phys.* **1984**, *36*, 831–841. [\[CrossRef\]](#)
- Bo-Ming, Y.; Kai-Lun, Y. Critical Percolation Probabilities for Site Problems on Sierpinski Carpets. *Z. Phys. B-Condens. Matter* **1988**, *70*, 209–212. [\[CrossRef\]](#)
- Lin, Z.Q.; Yang, Z.R. Thresholds and Universality of the Site Percolation on the Sierpinski Carpets. *Commun. Theor. Phys.* **1997**, *27*, 145–152. [\[CrossRef\]](#)
- Zhen-quan, L.; Zhan-ru, Y.; Yong, Q. Renormalization group approach to the bond percolation on Sierpinski carpets. *Acta Phys. Sin.* **1997**, *6*, 257–265. [\[CrossRef\]](#)
- Perreau, M.; Peiro, J.; Berthier, S. Percolation in random-Sierpinski carpets: A real space renormalization group approach. *Phys. Rev. E* **1996**, *54*, 4590–4595. [\[CrossRef\]](#) [\[PubMed\]](#)
- Hovi, J.-P.; Aharony, A. Renormalization group calculation of distribution functions: Structural properties for percolation clusters. *Phys. Rev. E* **1997**, *56*, 172–184. [\[CrossRef\]](#)
- Sukop, M.C.; Van Dijk, G.J.; Perfect, E.; Van Loon, W.K.P. Percolation Thresholds in 2-Dimensional Prefractal Models of Porous Media. *Transp. Porous Med.* **2002**, *48*, 187–208. [\[CrossRef\]](#)
- Corso, G.; Freitas, J.E.; Lucena, L.S.; Soares, R.F. Percolation on a multifractal. *Phys. Rev. E* **2004**, *69*, 066135. [\[CrossRef\]](#)
- Hassan, M.K.; Rahman, M.M. Percolation on a multifractal scale-free planar stochastic lattice and its universality class. *Phys. Rev. E* **2015**, *92*, 040101. [\[CrossRef\]](#)
- Hassan, M.K.; Rahman, M.M. Universality class of site and bond percolation on multifractal scale-free planar stochastic lattice. *Phys. Rev. E* **2016**, *94*, 042109. [\[CrossRef\]](#) [\[PubMed\]](#)
- Rahman, M.M.; Hassan, M.K. Explosive percolation on a scale-free multifractal weighted planar stochastic lattice. *Phys. Rev. E* **2017**, *95*, 042133. [\[CrossRef\]](#) [\[PubMed\]](#)



42. Herega, A.N.; Drik, N.G.; Ugol'nikov, A.P. Hybrid ramified Sierpinski carpet: Percolation transition, critical exponents, and force field. *Physics-Uspekhi* **2012**, *55*, 519–521. [\[CrossRef\]](#)
43. Wu, X.Y. Uniqueness of the Infinite Open Cluster for High-density Percolation on Lattice Sierpinski Carpet. *Acta Math. Sin.* **2001**, *17*, 141–146. [\[CrossRef\]](#)
44. Machlus, S.; Zhang, S.; Mao, X. Correlated rigidity percolation in fractal lattices. *Phys. Rev. E* **2011**, *103*, 012104. [\[CrossRef\]](#)
45. Liu, X.W.; Deng, Y.; Jacobsen, J.L. Recursive percolation. *Phys. Rev. E* **2015**, *92*, 010103. [\[CrossRef\]](#) [\[PubMed\]](#)
46. Shinoda, M. Existence of phase transition of percolation on Sierpinski carpet lattices. *J. Appl. Prob.* **2002**, *39*, 1–10. [\[CrossRef\]](#)
47. Monceau, P.; Yi Hsiao, P.Y. Percolation transition in fractal dimensions. *Phys. Lett. A* **2004**, *332*, 310–319. [\[CrossRef\]](#)
48. Higuchi, Y.; Wu, X.Y. Uniqueness of the critical probability for percolation in the two-dimensional Sierpinski carpet lattice. *Kobe J. Math.* **2008**, *25*, 1–24.
49. Balankin, A.S.; Martínez-Cruz, M.A.; Susarrey-Huerta, O.; Damian-Adame, L. Percolation on infinitely ramified fractal networks. *Phys. Lett. A* **2018**, *382*, 12–19. [\[CrossRef\]](#)
50. Balankin, A.S.; Martínez-Cruz, M.A.; Álvarez-Jasso, M.D.; Patiño-Ortiz, M.; Patiño-Ortiz, J. Effects of ramification and connectivity degree on site percolation threshold on regular lattices and fractal networks. *Phys. Lett. A* **2019**, *383*, 957–966. [\[CrossRef\]](#)
51. Halberstam, N.; Hutchcroft, T. What are the limits of universality? *Proc. Royal Soc. A* **2022**, *478*, 2259. [\[CrossRef\]](#)
52. Christensen, K. *Percolation Theory*; Imperial College London: London, UK, 2002.
53. Scullard, C.R. Exact site percolation thresholds using a site-to-bond transformation and the star-triangle transformation. *Phys. Rev. E* **2006**, *73*, 016107. [\[CrossRef\]](#)
54. Ziff, R.M.; Scullard, C.R. Exact bond percolation thresholds in two dimensions. *J. Phys. A Math. Gen.* **2006**, *39*, 15083–15090. [\[CrossRef\]](#)
55. Kownacki, J.-P. Site percolation on planar  $\Phi_3$  random graphs. *Phys. Rev. E* **2008**, *77*, 021121. [\[CrossRef\]](#) [\[PubMed\]](#)
56. Sykes, M.F.; Essam, J.W. Exact Critical Percolation Probabilities for Site and Bond Problems in Two Dimensions. *J. Math. Phys.* **1964**, *5*, 1117–1127. [\[CrossRef\]](#)
57. Becker, A.M.; Ziff, R.M. Percolation thresholds on two-dimensional Voronoi networks and Delaunay triangulations. *Phys. Rev. E* **2009**, *80*, 041101. [\[CrossRef\]](#) [\[PubMed\]](#)
58. Ziff, R.M.; Suding, P.N. Determination of the bond percolation threshold for the Kagomé lattice. *J. Phys. A Math. Gen.* **1997**, *30*, 5351–5359. [\[CrossRef\]](#)
59. Suding, P.N.; Ziff, R.M. Site percolation thresholds for Archimedean lattices. *Phys. Rev. E* **1999**, *60*, 275–283. [\[CrossRef\]](#)
60. Lorenz, C.D.; Ziff, R.M. Precise determination of the bond percolation thresholds and finite-size scaling corrections for the sc, fcc, and bcc lattices. *Phys. Rev. E* **1998**, *57*, 230–236. [\[CrossRef\]](#)
61. Van der Marck, S.C. Calculation of percolation thresholds in high dimensions for fcc, bcc and diamond lattices. *Int. J. Mod. Phys. C* **1998**, *9*, 529–540. [\[CrossRef\]](#)
62. Wang, J.; Zhou, Z.; Zhang, W.; Garoni, T.M.; Deng, Y. Bond and site percolation in three dimensions. *Phys. Rev. E* **2013**, *87*, 052107. [\[CrossRef\]](#)
63. Mertens, S.; Moore, C. Percolation thresholds and Fisher exponents in hypercubic lattices. *Phys. Rev. B* **2018**, *98*, 022120. [\[CrossRef\]](#)
64. Stauffer, D.; Jayaprakash, C. Critical exponents for one-dimensional percolation clusters. *Phys. Lett. A* **1978**, *64*, 433–434. [\[CrossRef\]](#)
65. Smirnov, S.; Werner, W. Critical exponents for two-dimensional percolation. *Math. Res. Lett.* **2001**, *8*, 729–744. [\[CrossRef\]](#)
66. Gracey, J.A. Four loop renormalization of  $\phi^3$  theory in six dimensions. *Phys. Rev. D* **2015**, *92*, 025012. [\[CrossRef\]](#)
67. Duminil-Copin, H.; Manolescu, I. Planar random-cluster model: Scaling relations. *Forum Math. Pi* **2022**, *10*, 1–83. [\[CrossRef\]](#)
68. Gwak, S.-H.; Goh, K.-I. No-exclaves percolation. *J. Korean Phys. Soc.* **2022**, *81*, 680–687. [\[CrossRef\]](#) [\[PubMed\]](#)
69. Stanley, N.E. Application of fractal concepts to polymer statistics and to anomalous transport in randomly porous media. *J. Stat. Phys.* **1984**, *36*, 843–860. [\[CrossRef\]](#)
70. Stella, A.L.; Vanderzande, C. Scaling and fractal dimension of Ising clusters at the d-2 critical point. *Phys. Rev. Lett.* **1989**, *962*, 1067–1070. [\[CrossRef\]](#) [\[PubMed\]](#)
71. Higuchi, Y.; Takei, M.; Zhang, Y. Scaling relations for two-dimensional Ising percolation. *J. Stat. Phys.* **2012**, *148*, 777–799. [\[CrossRef\]](#)
72. Balankin, A.S.; Martínez-Cruz, M.A.; Gayosso-Martínez, F.; Mena, B.; Tobon, A.; Patiño-Ortiz, J.; Patiño-Ortiz, M.; Samayoa, D. Ising percolation in a three-state majority vote model. *Phys. Lett. A* **2017**, *381*, 440–445. [\[CrossRef\]](#)
73. Kutlu, B. Critical exponents of the two-dimensional Ising model with next-nearest-neighbor and four-spin interactions on the Creutz cellular automaton. *Phys. A* **1997**, *243*, 199–212. [\[CrossRef\]](#)
74. Pelissetto, A.; Vicari, E. Critical phenomena and renormalization-group theory. *Phys. Rep.* **2002**, *368*, 549–727. [\[CrossRef\]](#)
75. Wipf, A. Classical Spin Models: An Introduction. In *Statistical Approach to Quantum Field Theory, Lecture Notes in Physics*; Springer: Cham, Switzerland, 2021; Volume 992. [\[CrossRef\]](#)
76. Fytas, N.G.; Martín-Mayor, V.; Picco, M.; Sourlas, N. Restoration of dimensional reduction in the random-field Ising model at five dimensions. *Phys. Rev. E* **2017**, *95*, 042117. [\[CrossRef\]](#) [\[PubMed\]](#)
77. Kumar, M.; Banerjee, V.; Puri, S.; Weigel, M. Critical behavior of the three-state random-field Potts model in three dimensions. *arXiv* **2022**, arXiv:2205.13047. [\[CrossRef\]](#)
78. Holm, C.; Janke, W. Critical exponents of the classical three-dimensional Heisenberg model: A single-cluster Monte Carlo study. *Phys. Rev. B* **1993**, *48*, 936–950. [\[CrossRef\]](#)
79. Campostrini, M.; Hasenbusch, M.; Pelissetto, A.; Rossi, P.; Vicari, E. Critical behavior of the three-dimensional XY universality class. *Phys. Rev. B* **2001**, *63*, 214503. [\[CrossRef\]](#)
80. Goldenfeld, N. *Lectures on Phase Transitions and the Renormalization Group*; CRC Press: Boca Raton, FL, USA, 1992. [\[CrossRef\]](#)

81. Qing, Y.; You, W.L.; Liu, M. Critical exponents and the universality class of a minesweeper percolation model. *Int. J. Mod. Phys. C* **2020**, *31*, 2050129. [[CrossRef](#)]
82. Balankin, A.S. A continuum framework for mechanics of fractal materials I: From fractional space to continuum with fractal metric. *Eur. Phys. J. B* **2015**, *88*, 90. [[CrossRef](#)]
83. Balankin, A.S. Fractional space approach to studies of physical phenomena on fractals and in confined low-dimensional systems. *Chaos Solitons Fractals* **2020**, *132*, 109572. [[CrossRef](#)]
84. Balankin, A.S.; Patiño-Ortiz, J.; Patiño-Ortiz, M. Inherent features of fractal sets and key attributes of fractal models. *Fractals* **2022**, *30*, 2250082. [[CrossRef](#)]
85. Balankin, A.S. The topological Hausdorff dimension and transport properties of Sierpinski carpets. *Phys. Lett. A* **2017**, *381*, 2801–2808. [[CrossRef](#)]
86. Balankin, A.S.; Mena, B.; Martinez-Cruz, M.A. Topological Hausdorff dimension and geodesic metric of critical percolation cluster in two dimensions. *Phys. Lett. A* **2017**, *381*, 2665–2672. [[CrossRef](#)]
87. Balankin, A.S. Effective degrees of freedom of a random walk on a fractal. *Phys. Rev. E* **2015**, *92*, 062146. [[CrossRef](#)]
88. Havlin, S.; Ben-Avraham, D. Diffusion in disordered media. *Adv. Phys.* **2002**, *51*, 187–292. [[CrossRef](#)]
89. Burioni, R.; Cassi, D. Random walks on graphs: Ideas, techniques and results. *J. Phys. A Math. Gen.* **2005**, *38*, R45–R78. [[CrossRef](#)]
90. Balankin, A.S.; Golmankhaneh, A.K.; Patiño-Ortiz, J.; Patiño-Ortiz, M. Noteworthy fractal features and transport properties of Cantor tartans. *Phys. Lett. A* **2018**, *382*, 1534–1539. [[CrossRef](#)]
91. Durhuus, B. Hausdorff and spectral dimension of infinite random graphs. *Acta Phys. Pol. B* **2009**, *40*, 3509–3532. Available online: <https://s3.cern.ch/inspire-prod-files-a/a84919e9e7db6236283e133b0dc0314a> (accessed on 7 January 2023).
92. Balankin, A.S.; Ramírez-Joachin, J.; Gonzalez-Lopez, G.; Gutiérrez-Hernandez, S. Formation factors for a class of deterministic models of pre-fractal pore-fracture networks. *Chaos Solitons Fractals* **2022**, *162*, 112452. [[CrossRef](#)]
93. March, N.H. Toward a final theory of critical exponents in terms of dimensionality  $d$  plus universality class  $n$ . *Phys. Lett. A* **2015**, *379*, 820–822. [[CrossRef](#)]
94. Bonnier, B.; Leroyer, Y.; Meyers, C. Critical exponents for Ising-like systems on Sierpinski carpets. *J. Phys.* **1987**, *48*, 553–558. [[CrossRef](#)]
95. Le Guillou, J.C.; Zinn-Justin, J. Accurate critical exponents for Ising like systems in non-integer dimensions. *J. Phys.* **1987**, *48*, 19–24. [[CrossRef](#)]
96. Holovatch, Y.; Shpot, M. Critical Exponents of Random Ising-Like Systems in General Dimensions. *J. Stat. Phys.* **1992**, *66*, 867–883. [[CrossRef](#)]
97. Monceau, P.; Perreau, M. Critical behavior of the Ising model on fractal structures in dimensions between one and two: Finite-size scaling effects. *Phys. Rev. B* **2001**, *63*, 184420. [[CrossRef](#)]
98. Bab, M.A.; Fabricius, G.; Albano, E.V. Critical behavior of an Ising system on the Sierpinski carpet: A short-time dynamics study. *Phys. Rev. E* **2005**, *71*, 036139. [[CrossRef](#)]
99. Bab, M.A.; Fabricius, G.; Albano, E.V. Critical exponents of the Ising model on low-dimensional fractal media. *Phys. A* **2009**, *388*, 370–378. [[CrossRef](#)]
100. Kazmin, S.; Janke, W. Critical exponents of the Ising model in three dimensions with long-range power-law correlated site disorder: A Monte Carlo study. *Phys. Rev. B* **2022**, *105*, 214111. [[CrossRef](#)]
101. Newman, M.E.J.; Ziff, R.M. Efficient Monte Carlo algorithm and high-precision results for percolation. *Phys. Rev. Lett.* **2000**, *85*, 4104–4107. [[CrossRef](#)]
102. Sahimi, M.; Rassamdana, H. On position-space renormalization group approach to percolation. *J. Stat. Phys.* **1995**, *78*, 1157–1164. [[CrossRef](#)]
103. Wierman, J.C.; Naor, D.P. Criteria for evaluation of universal formulas for percolation thresholds. *Phys. Rev. E* **2005**, *71*, 036143. [[CrossRef](#)] [[PubMed](#)]
104. Wierman, J.C. Accuracy of universal formulas for percolation thresholds based on dimension and coordination number. *Phys. Rev. E* **2002**, *66*, 027105. [[CrossRef](#)]
105. Fisher, M.E. Critical probabilities for cluster size and percolation problems. *J. Math. Phys.* **1961**, *2*, 620–627. [[CrossRef](#)]
106. van den Berg, J. Percolation theory on pairs of matching lattices. *J. Math. Phys.* **1981**, *22*, 152–157. [[CrossRef](#)]
107. Kesten, H. *Percolation Theory for Mathematicians*; Birkhäuser: Boston, MA, USA, 1982.
108. Ohzeki, M.; Fujii, K. Duality analysis on random planar lattices. *Phys. Rev. E* **2012**, *86*, 051121. [[CrossRef](#)] [[PubMed](#)]
109. Wierman, J.C. Bond percolation critical probability bounds derived by edge contraction. *J. Phys. A* **1988**, *21*, 1487–1492. [[CrossRef](#)]
110. Riordan, O.; Walters, M. Rigorous confidence intervals for critical probabilities. *Phys. Rev. E* **2007**, *76*, 011110. [[CrossRef](#)]
111. Scullard, C.R.; Ziff, R.M. Critical surfaces for general bond percolation problems. *Phys. Rev. Lett.* **2008**, *100*, 185701. [[CrossRef](#)]
112. Ball, N. Rigorous confidence intervals on critical thresholds in 3 dimensions. *J. Stat. Phys.* **2014**, *156*, 574–585. [[CrossRef](#)]
113. Teixeira, A. Percolation and local isoperimetric inequalities. *Probab. Theory Relat. Fields* **2016**, *165*, 963–984. [[CrossRef](#)]
114. Wierman, J.C. On bond percolation threshold bounds for Archimedean lattices with degree three. *J. Phys. A Math. Theor.* **2017**, *50*, 295001. [[CrossRef](#)]
115. Torquato, S.; Jiao, Y. Effect of dimensionality on the percolation thresholds of various  $d$ -dimensional lattices. *Phys. Rev. E* **2013**, *87*, 032149. [[CrossRef](#)]

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